04/16/2026: Coding Workshop: Attention

Reference: *Dive Into Deep Learning*, Chapter 10.

CSCI 546: Diffusion Models

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Course Logistics

Feedback on Reading Survey #8

The original question was: What is attention, and how does it work?

In general, people provided great answers to the first question, but many people struggled (or didn't answer) the second question.

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In general, people provided great answers to the first question, but many people struggled (or didn't answer) the second question.

The easiest way to answer the second question is to reference the notion of keys, queries, and values.

Poll. Would anybody like to give a verbal answer to the second question?

Coding Workshop

Mini-Lecture

Attention as Nadaraya-Watson Kernel Regression

Nonparametric Regression

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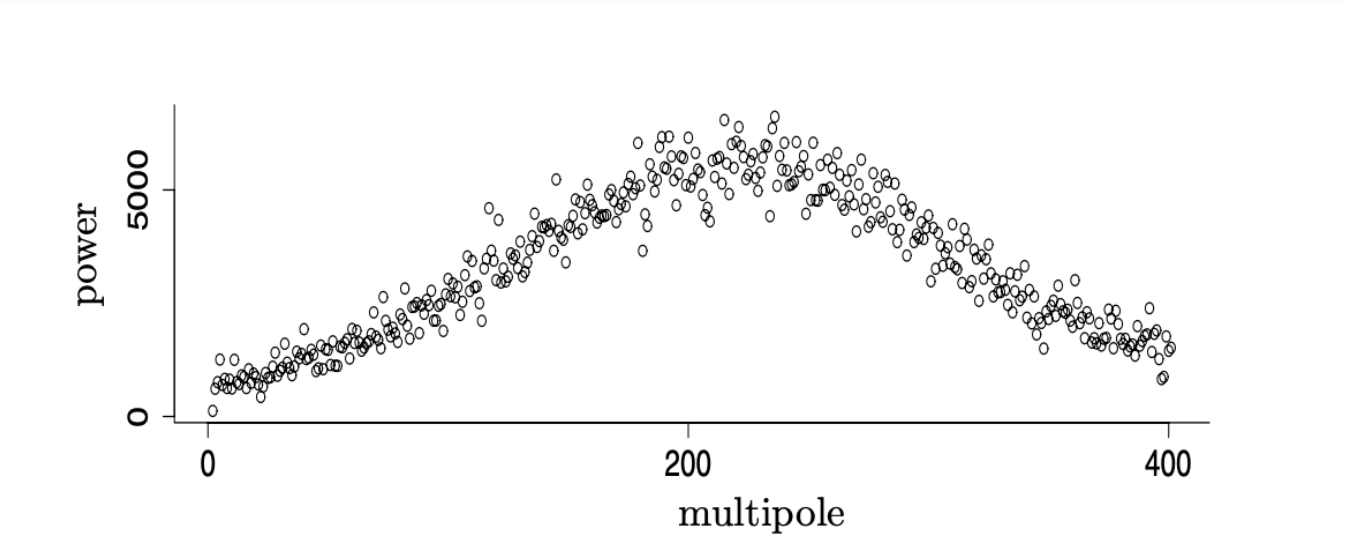
This allows us to perform regressions for datasets such as the Cosmic Microwave Background (CMB) radiation:



Nonparametric Regression

We assume the **response variable** Y is related to known **covariates** $X = x$ by the equation

$$Y_i = r(x_i) + \epsilon_i, \qquad \mathbb{E}[\epsilon_i] = 0, \quad i = 1, \ldots, n.$$

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Where we interpret the "trend" $r(x)$ as the mean of Y conditional on $X = x$, that is

$$r(x) = \mathbb{E}[Y \mid X = x].$$

Smoothing Kernel

Smoothing kernels are for taking local averages.

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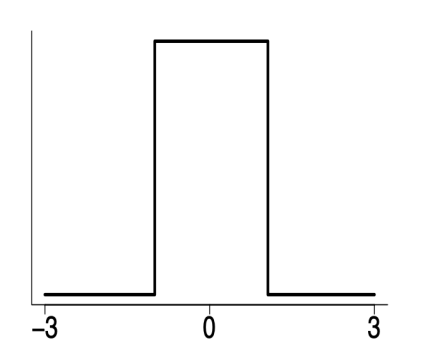
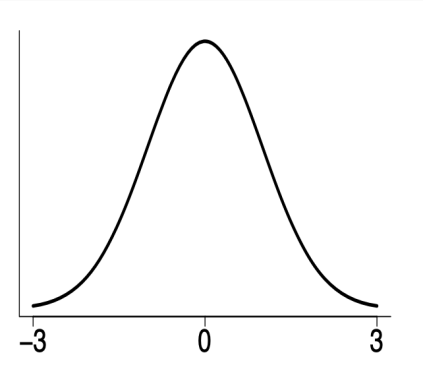
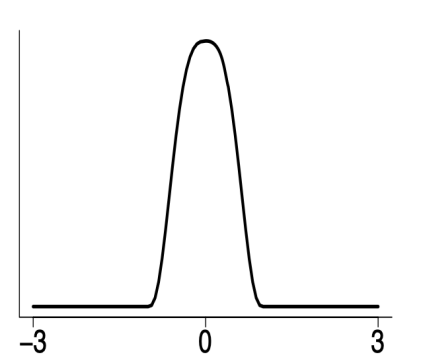


Boxcar



Gaussian

Epanechnikov



Tricube

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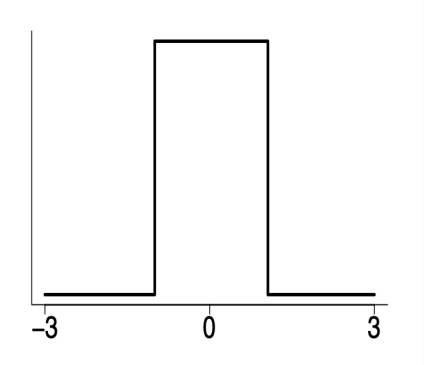
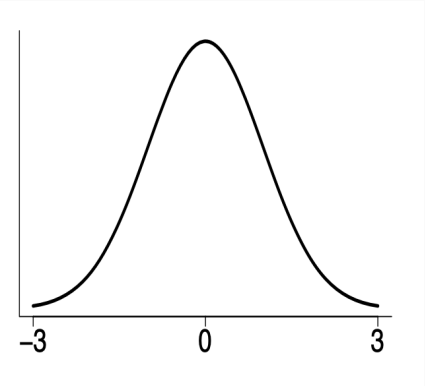
Smoothing Kernel



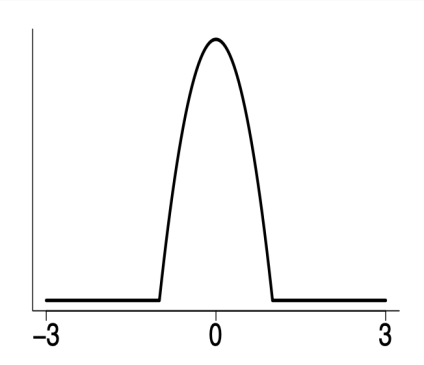
Boxcar

$K(x) = \frac{1}{2}\, I(x)$



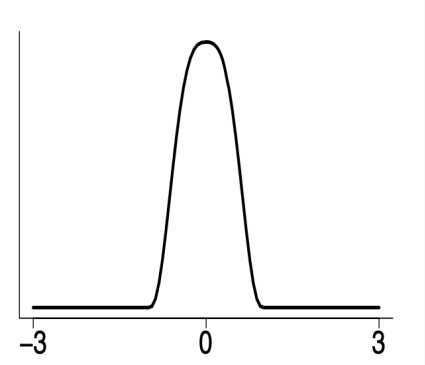
Gaussian

$K(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$



Epanechnikov

$K(x) = \frac{3}{4}(1 - x^2)\, I(x)$



Tricube

$K(x) = \frac{70}{81}(1 - |x|^3)^3\, I(x)$

$$I(x) = \begin{cases} 1, & \text{if } |x| \leq 1 \\ 0, & \text{if } |x| > 1 \end{cases}$$

Smoothing Kernel

Definition

A **smoothing kernel** is any smooth function K such that $K(x) > 0$ and

$$\int K(x)\mathrm{d}x = 1, \qquad \int x\,K(x)\mathrm{d}x = 0, \quad \text{and} \quad \sigma_K^2 := \int x^2 K(x)\mathrm{d}x > 0$$

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Poll. What's an easy way to construct a smoothing kernel?

Solution. Just take the p.d.f. of a (non-degenerate) random variable with mean 0.

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Estimating the trend

Definition

Let $h > 0$ be a positive number, called the **bandwidth**. The **Nadaraya-Watson kernel estimator** is defined by

$$\widehat{r}(x) = \sum_{i=1}^{n} \ell_i(x) y_i$$

where K is a smoothing kernel and the weights $\ell_i(x)$ are given by

$$\ell_i(x) = \frac{K(\frac{x-x_i}{h})}{\sum_{j=1}^{n} K(\frac{x-x_j}{h})} \tag{1}$$

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Basically, we estimate the trend $r(x)$ by taking a weighted average of the y_i's, giving higher weight to those points that are "close" to x.

Example: CMB Data

Let's use Nadayara-Watson kernel regression to estimate the trend for CMB data.

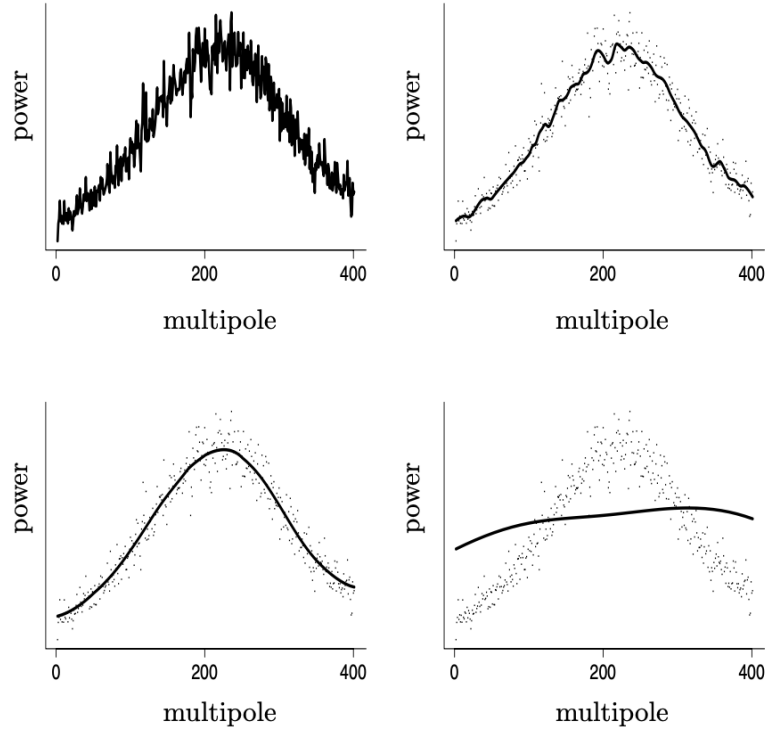

FIGURE 5.2. Four kernel regressions for the CMB data using just the first 400 data points. The bandwidths used were $h = 1$ (top left), $h = 10$ (top right), $h = 50$ (bottom left), $h = 200$ (bottom right). As the bandwidth h increases, the estimated function goes from being too rough to too smooth.

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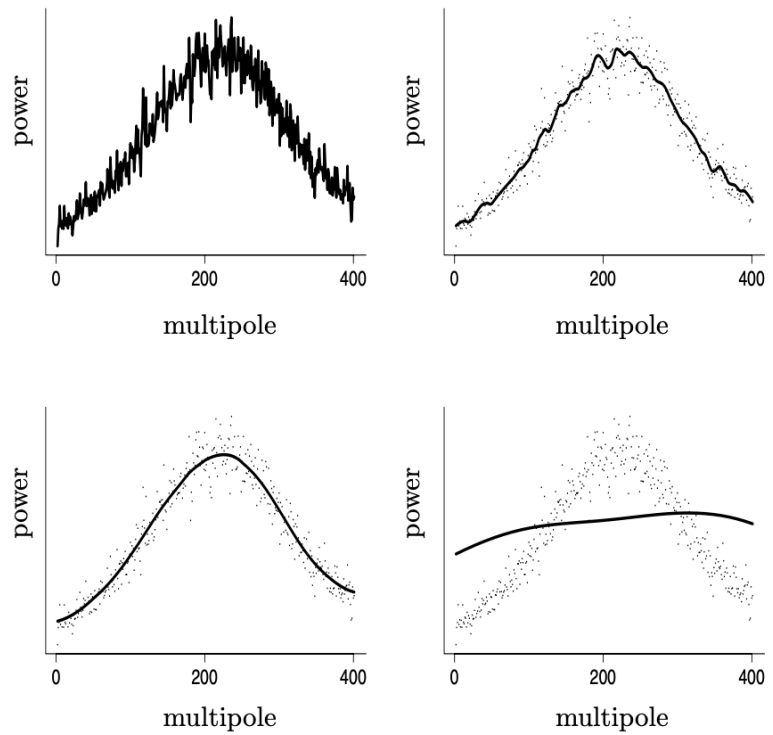

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Remark. The optimal bandwidth can be chosen via cross-validation.

Discussion: How is attention pooling related to Nadayara-Watson kernel regression?

Summary

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This kernel is learned rather than fixed in advance.